

## **Summary of AI Applications**

For the present research study, several applications of AI technology have been applied for carrying out different tasks related to the research, including modeling, programming, and reviewing literature, among others. All results achieved were carefully reviewed and edited by the author.

## **List of Used AI and Their Tasks**

- Gemini: Created research diagrams illustrating the entire process required to develop a model and carry out research.
- ChatGPT: Suggested models that can be employed during the course of research; analyzed and debugged code; provided summary of major research findings.
- Perplexity AI: Conducted literature review to come up with research findings and trends; identified gaps within the previous research.
- NotebookLM: Helped in quickly reviewing literature, as well as creating summaries and outlining methods and results used in research.
- Claude: Developed the code needed for the implementation of suggested model.

## **Workflow Steps**

- Literature Review: Perplexity AI, NotebookLM, and ChatGPT
- Model Development: ChatGPT and Gemini
- Visualization: Gemini
- Code Generation: Claude
- Explanations and Debugging: ChatGPT

## **Human Input**

All the outcomes of AI usage were carefully reviewed, edited, and validated by the author. The researcher made the final determination on the most appropriate model and its outcomes.

## **Conclusion**

The tools and techniques discussed above played a crucial role in fast-tracking research tasks and activities; nonetheless, scientific reasoning and judgments remain human-oriented.

